

Maritime Oil Spill Forensic Investigation Report

REVERSE-LAGRANGIAN HYDRODYNAMICS & AIS ATTRIBUTION FORENSIC DOSSIER

Target Incident:
TAYLOR_ENERGY

Analysis Date: 2026-09-26

Investigation Standard: IMO & NOAA Damage Assessment Protocol

PRIMARY VERDICT
SUSPECTED_ANCHOR_STRIKE_ON_PIPELINE

ATTRIBUTION
CONFIDENCE
HIGH

ORIGIN ERROR
(GROUND TRUTH)
309 m

VESSELS
EXONERATED
15 / 15

Official Forensic Determination

Executive Finding: Vessel CG WALNUT (MMSI: 366953000) was observed loitering/anchoring within 119m of Pipeline Segment 12712 operated by WALTER OIL & GAS CORPORATION prior to release.

Spill Origin Coordinates: 28.93537°N, -88.96780°W | **Estimated Release Window:** 2023-11-17T23:16:46+00:00 to 2023-11-17T23:24:16+00:00

Scientific Benchmark Ground Truth Verification

Known Site: Taylor Energy MC-20 Wellhead Site (BSEE / NOAA Official)

Official Coordinates: 28.93653°N, -88.97069°W

Model Reconstructed Origin: 28.93537°N, -88.96780°W

Absolute Geodetic Error: 309.4 meters (0.167 nautical miles)

Hurricane Ivan (Sept 2004) triggered an underwater mudslide that toppled the Taylor Energy Mississippi Canyon Block 20 platform and buried 28 wellheads. Continuous active containment system operated by Couvillion Group under USCG/BSEE oversight.

1. SATELLITE REMOTE SENSING & SLICK MORPHOLOGY

Morphological Metric	Calculated Value	Forensic Significance
Satellite Overpass Time	2023-11-17T23:54:16+00:00	Sentinel-1 Synthetic Aperture Radar (SAR) acquisition
Observed Slick Footprint	28.93471°N, -88.96548°W	Surface polygon center of mass
Major Spreading Axis Length	2.67 km	Linear extent along principal dispersion direction
Transverse Dimensions	Head: 1780 m Tail: 2136 m	Diffusion profile: $W_{tail} > W_{head}$ (widening downstream profile confirms active drift from source)
Calculated Drift Age (ΔT)	0.50 hours	Fickian turbulent diffusion model (turbulent diffusivity $K_h = 10 \text{ m}^2/\text{s}$)

Diffusion Age Formulation: $\Delta T = (W_{\text{tail}}^2 - W_{\text{head}}^2) \div (8 \times K_h)$

Substituting $W_{\text{tail}} = 2136.1$ m, $W_{\text{head}} = 1780.1$ m, and $K_h = 10.0$ m²/s yields an elapsed residence time of **0.50 hours**.

2. HYDRODYNAMIC BACKTRACK SIMULATION (OPENDRIFT UPSTREAM)

OpenDrift 1.14 reverse-Lagrangian trajectory simulation seeded with 500 particles, driven by Copernicus Marine Service (CMEMS) reanalysis currents, 10m winds, and Stokes wave drift.

Physical Forcing Field	Source Model	Resolution & Parameterization
Ocean Surface Currents	CMEMS Global Analysis	Hourly zonal (u_o) and meridional (v_o) surface vectors
Stokes Wave Drift	CMEMS Global Wave Model	Wave radiation stress vectors ($VSDX$, $VSDY$)
Atmospheric Wind Leeway	Copernicus / ECMWF 10m	3.0% leeway coefficient with 5° Coriolis deflection
Sub-grid Dispersion	Lagrangian Random Walk	Horizontal turbulent diffusivity $K_h = 10.0$ m ² /s

3. SUBSEA & OFFSHORE INFRASTRUCTURE ASSESSMENT

Cross-referenced against official BOEM/BSEE deepwater pipeline and platform spatial datasets:

Infrastructure Parameter	Observed Record	Regulatory Context
Nearest Asset	Subsea Pipeline	BSEE / BOEM Marine Cadastre
Distance to Spill Origin	49.1 meters	Co-located within physical dispersion boundary
Segment / Asset ID	12712	Operator: WALTER OIL & GAS CORPORATION
Diameter / Status	06" / Status: ABN	Product Code: BLKG

4. MARITIME TRAFFIC & CANDIDATE VESSEL EVALUATION (AIS DOWNSTREAM)

Vessels within the 4D spatio-temporal query corridor resampled at 1-minute intervals for Closest Point of Approach (CPA):

#	Vessel Name	MMSI	Classification	CPA Dist	CPA Time	SOG	Score	Level
1	CG WALNUT	366953000	Offshore / Tug / Other (90)	8451 m	23:18:55	13.3 kt	52.5%	MEDIUM
2	KARLA F	367332840	Offshore / Tug / Other (90)	10915 m	23:44:30	9.8 kt	51.7%	MEDIUM
3	MATTERHORN TLP	367376530	Offshore / Tug / Other (90)	25473 m	23:30:19	0.1 kt	50.3%	MEDIUM
4	MR SEAMAN	367154130	Passenger (60)	10723 m	23:40:24	1.9 kt	47.5%	LOW
5	FULL CIRCLE II	338444731	Fishing / Trawler (37)	31966 m	22:59:59	15.2 kt	43.7%	LOW
6	GO GLORY	368269780	Cargo (70)	14243 m	23:02:25	0.1 kt	43.1%	LOW
7	NGUYEN T J	367594140	Fishing / Trawler (30)	14976 m	23:58:45	7.6 kt	42.7%	LOW
8	MISSISSIPPI III	367724890	Fishing / Trawler (30)	16704 m	23:17:50	7.6 kt	41.0%	LOW
9	BRETON ISLAND	367558730	Offshore / Tug / Other (90)	13755 m	22:17:14	2.5 kt	39.9%	LOW
10	BOOGIE	338176375	Fishing / Trawler (37)	33202 m	22:58:06	28.5 kt	39.8%	LOW
11	KIMBERLY CELESTE	368152930	Fishing / Trawler (30)	14213 m	22:17:19	6.0 kt	38.6%	LOW
12	JAMAL	366701330	Fishing / Trawler (37)	33731 m	22:59:38	0.1 kt	37.6%	LOW

5. MULTI-CRITERIA DECISION ANALYSIS (MCDA) EXONERATION MATRIX

Exoneration Justification: All detected commercial vessels maintained a minimum CPA clearance of >8.4 km from the physical release origin. Their low suspicion scores (31.1% – 52.5%) combined with high spatial separation categorically **clears all passing surface traffic of responsibility**.

The physical origin coincides within 310 meters of the known Taylor Energy MC-20 wellhead structure, verifying that the observed discharge is an **ongoing infrastructure failure** rather than an illicit vessel discharge.